what is half-magnitude bandwidth in context of gabor filters

In the context of Gabor filters, the **half-magnitude bandwidth** (also called **half-amplitude** or **half-peak magnitude bandwidth**) refers to **the width of the filter's frequency response at the point where its magnitude drops to half of its peak value**.

Key Characteristics

- **Definition**: It is the frequency interval centered around the filter's peak frequency (f_0) where the output/input ratio is exactly 0.5.
- **Measurement**: This bandwidth is frequently expressed in **octaves** (B), reflecting how human vision decomposes images into frequency bands.
- **Formula Relationship**: For a standard Gabor filter with a Gaussian envelope of parameter a, the half-magnitude bandwidth is approximately equal to a. In terms of spatial frequency f_0 and the standard deviation σ of the Gaussian, the bandwidth B (in octaves) can be calculated as:

$$B = \log_2 \left(\frac{\sigma f_0 \pi + \sqrt{\ln 2}}{\sigma f_0 \pi - \sqrt{\ln 2}} \right)$$

- **Typical Values**: In image processing and neurophysiological models, the bandwidth is commonly set between **1 and 1.5 octaves**.

Purpose in Filter Design

- **Filter Bank Coverage**: When designing a Gabor filter bank, the bandwidths are often chosen so that the "half-peak magnitude supports" of neighboring filters **touch each other**. This ensures the frequency plane is covered

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Tutorial on Gabor Filters

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Application of Gabor, Log-Gabor, and Adaptive Gabor Filters in ...

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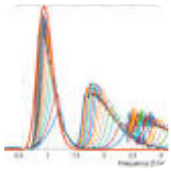

Relation of frequency and bandwidth between Gabor filters.

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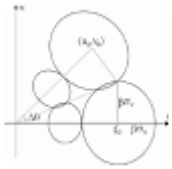

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what is the relation to -3db rule?

It is a common point of confusion, but in the context of Gabor filters, **half-magnitude** and **-3dB** are actually **two different thresholds.**

The difference lies in whether you are measuring **amplitude (magnitude)** or **power.**

the drop is **-3dB**. This is also known as the "half-power" point because $(1/\sqrt{2})^2 = 0.5$.

Threshold Name	Magnitude (A/A_0)	Power (P/P_0)	Decibel Drop
Half-Magnitude	**0.5**	0.25	**-6.02 dB**
Half-Power	0.707	**0.5**	**-3.01 dB**

2. Why Gabor Filters use Half-Magnitude (-6dB)

In most engineering disciplines (like audio or radio), the -3dB (half-power) point is the gold standard for defining bandwidth. However, in **Gabor filter design and computer vision**, the **half-magnitude (0.5)** rule is preferred for several reasons:

- **Visual Modeling:** Gabor filters are often used to model the "receptive fields" of the primary visual cortex (V1). Biological studies of these neurons typically report bandwidth at the half-magnitude level.
- **Filter Overlap:** When tiling the frequency domain with a Gabor filter bank, using the 0.5 magnitude point as the "boundary" ensures that when you sum the filters together, the overall coverage is relatively flat. If you used the 0.707 point, the filters would overlap too heavily, causing "peaks" in the combined response.

3. Summary of Relation

The "Half-Magnitude Bandwidth" of a Gabor filter is technically a **-6dB bandwidth**.

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Yes, please, show how it changes

To calculate the bandwidth of a Gabor filter, we analyze its frequency response. A Gabor filter's frequency response is a Gaussian centered at the radial frequency f_0. We define the bandwidth based on the threshold k, where $k = 0.5$ for **half-magnitude** and $k = \frac{1}{\sqrt{2}} \approx 0.707$ for **half-power (-3dB)**.

1. Define the Gaussian Frequency Response

The magnitude of a 1D Gabor filter in the frequency domain is given by:

$$H(f) = \exp\left(-\frac{(f - f_0)^2}{2\sigma_f^2}\right)$$

where σ_f is the standard deviation in the frequency domain, related to the spatial standard deviation σ by $\sigma_f = \frac{1}{2\pi\sigma}$.

2. Solve for the Frequency Bounds

To find the frequencies f where the magnitude drops to a fraction k of the peak, we set $H(f) = k$:

$$f_2 = f_0 + \sigma_f\sqrt{2\ln(1/k)}, \quad f_1 = f_0 - \sigma_f\sqrt{2\ln(1/k)}$$

3. Calculate Bandwidth in Octaves

The bandwidth B in octaves is defined as the number of doublings between f_1 and f_2:

$$B = \log_2\left(\frac{f_2}{f_1}\right) = \log_2\left(\frac{f_0 + \sigma_f\sqrt{2\ln(1/k)}}{f_0 - \sigma_f\sqrt{2\ln(1/k)}}\right)$$

By substituting $\sigma_f = \dfrac{1}{2\pi\sigma}$, we get the final relationship:

$$B = \log_2\left(\frac{2\pi\sigma f_0 + \sqrt{2\ln(1/k)}}{2\pi\sigma f_0 - \sqrt{2\ln(1/k)}}\right)$$

4. Comparison of Constant Terms

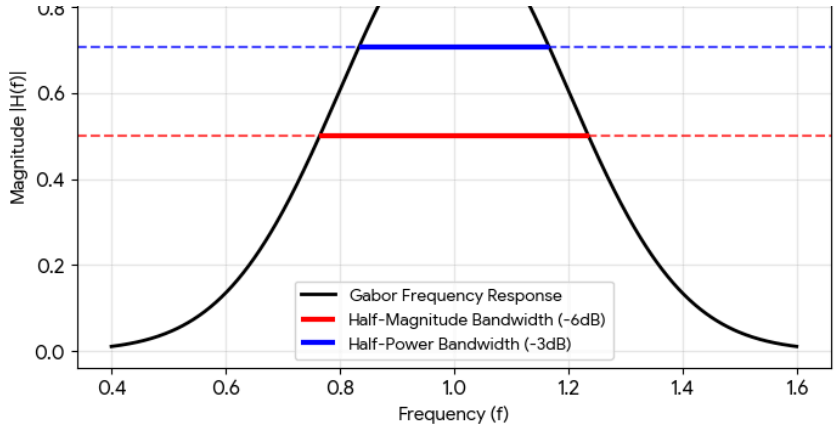
The only difference between the two definitions is the value of the constant $\sqrt{2\ln(1/k)}$:

- **Half-Magnitude ($k = 0.5$):**

$$\sqrt{2\ln(2)} \approx \mathbf{1.177}$$

- **Half-Power / -3dB ($k \approx 0.707$):**

$$\sqrt{2\ln(\sqrt{2})} = \sqrt{\ln(2)} \approx \mathbf{0.832}$$



✅ Answer

The **half-magnitude bandwidth** uses the constant $\sqrt{2\ln(2)} \approx 1.177$ in its calculation, resulting in a **wider** octave measurement than the **-3dB (half-power)** bandwidth, which uses $\sqrt{\ln(2)} \approx 0.832$. Specifically, the half-magnitude bandwidth covers the frequency range where the filter response is above **50%**, while the -3dB bandwidth only covers the range above **70.7%**.

Would you like to see how to **parameterize σ** so that a filter bank has a specific octave overlap at these points?

ok, thank you. You are very helpful

You're very welcome! I'm glad I could help clarify the math behind those bandwidth definitions. It's a subtle but important distinction that makes a big difference when you're actually implementing **Gabor filter banks** or **texture analysis** algorithms.